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Subject : DSA

Unit 3, S4, SLO1 - Stack Applications - Reversing a List and Balancing Symbols

1. Consider any 5 elements in a List in an array A = [10, 20, 30, 40, 50]. Use a Stack S using array implementation and Reverse the Elements of the List into array B.

Step 1: Representation of elements in the List

Step 2: Traverse list from first element and push into Stack S

Step 3: Pop elements one by one from TOP and place in array B

Answer: Step 1: Array A = [10, 20, 30, 40, 50], N = 5.

Step 2: Traverse A from index 0 to 4 and push each element onto Stack S:

- push(10) -> S: [10] (TOP=0)
- push(20) -> S: [10, 20] (TOP=1)
- push(30) -> S: [10, 20, 30] (TOP=2)
- push(40) -> S: [10, 20, 30, 40] (TOP=3)
- push(50) -> S: [10, 20, 30, 40, 50] (TOP=4)

Step 3: Pop elements from Stack S and place into array B:

- pop() -> 50, B[0] = 50
- pop() -> 40, B[1] = 40
- pop() -> 30, B[2] = 30
- pop() -> 20, B[3] = 20
- pop() -> 10, B[4] = 10

Result: Array B = [50, 40, 30, 20, 10] (Successfully Reversed!)

2. Check if the following expressions have balanced parenthesis using stack:

a. { P + (Q - R) }

b. { D * (S + A) }

Answer: a. Expression: { P + (Q - R) }

Step 1: Initialize flag = 1, Stack S = empty

Step 2: Scan '{' -> Push '{' onto stack S -> S = ['{']

Step 3: Scan 'P', '+' -> Operands/operators ignored

Step 4: Scan '(' -> Push '(' onto stack S -> S = ['{', '(']

Step 5: Scan 'Q', '-', 'R' -> Ignored

Step 6: Scan ')' -> Pop from S -> popped '(' matches ')' -> S = ['{']

Step 7: Scan '}' -> Pop from S -> popped '{' matches '}' -> S = empty

Step 8: End of expression. Stack is empty and flag = 1 -> VALID (Balanced Parentheses)

b. Expression: { D * (S + A) }

Step 1: Initialize flag = 1, Stack S = empty

Step 2: Scan '{' -> Push '{' -> S = ['{']

Step 3: Scan '(' -> Push '(' -> S = ['{', '(']

Step 4: Scan ')' -> Pop '(' -> S = ['{']

Step 5: Scan '}' -> Pop '{' -> Mismatched opening bracket ('{' does not match '}') -> flag = 0

Step 6: End of expression. flag = 0 -> INVALID (Unbalanced Parentheses)